approaches within the quantum-gravity community is to ask how we should think about the conceptual side of things. A string theorist is likely to believe that if we take care of all the technical issues, the conceptual problems will eventually resolve themselves. Someone who thinks otherwise might be nudged toward loop quantum gravity or another alternative approach. When the data don't point one way or the other, opinions tend to become deeply entrenched.

String theory, loop quantum gravity, and other ideas share a common pattern: they start with a set of classical variables, then quantize. From the perspective we've been following in this book, that's a little backward. Nature is quantum from the start, described by a wave function evolving according to an appropriate version of the Schrödinger equation. Things like "space" and "fields" and "particles" are useful ways of talking about that wave function in an appropriate classical limit. We don't want to start with space and fields and quantize them; we want to extract them from an intrinsically quantum wave function.

How can we find "space" within a wave function? We want to identify features of the wave function that resemble space as we know it, and in particular something that would correspond to a metric that defines distances. So let's think about how distances show up in ordinary quantum field theory. For simplicity, let's just think about distances in space; we'll talk later about how time might enter into the game.

There's one obvious place that distances show up in quantum field theory, which we've seen in the last chapter: in empty space, fields in different regions are entangled with each other, and regions that are far away are less entangled than ones that are nearby. Unlike "space," the concept of "entanglement" is always available to us in any abstract quantum wave function. So perhaps we can get some purchase here, looking at the entanglement structure of states and using that to define distances. What we need is a quantitative measure of how entangled a quantum subsystem actually is. Happily, such a measure exists: it's the entropy.

John von Neumann showed how quantum mechanics introduces a notion of entropy that parallels the classical definition. As explained by